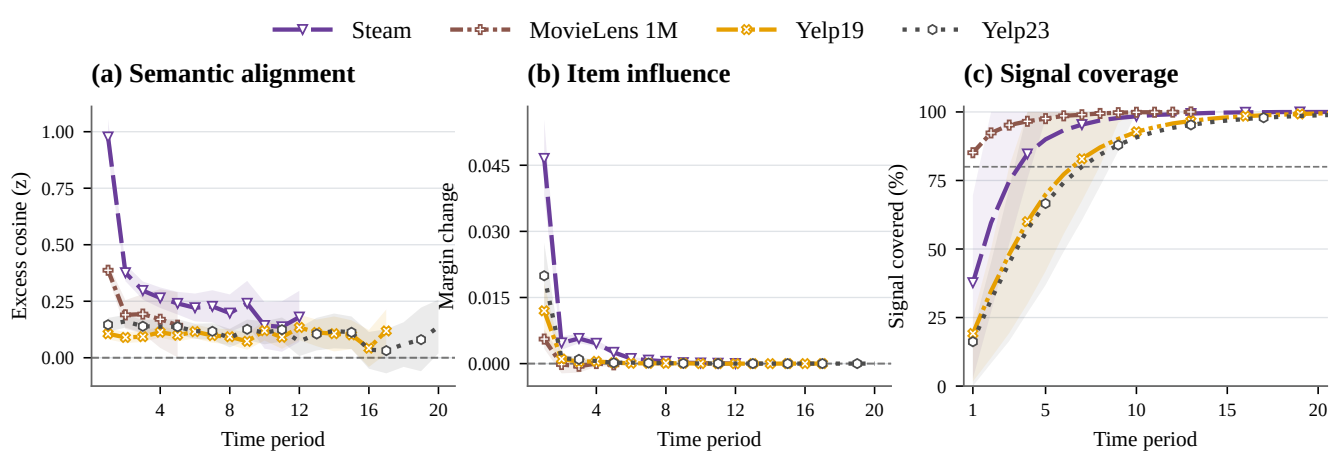
Steam
MovieLens 1M
Yelp19
Yelp23
(a) Semantic alignment
Excess cosine (z)
1.00
0.75
0.50
0.25
0.00
4
8
12
16
20
Time period
(b) Item influence
Margin change
0.045
0.030
0.015
0.000
4
8
12
16
20
Time period
(c) Signal coverage
Signal covered (%)
100
75
50
25
0
1
5
10
15
20
Time period